

1.1. Exponential increasing.

$$\varphi: \mathbb{R} \rightarrow \mathbb{R}, \varphi(t) = 2 \cdot e^{3t}, t \in \mathbb{R}.$$

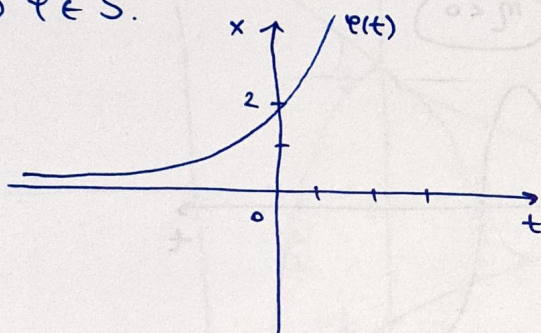
Check if φ is a solution of the IVP: $x' = 3x, x(0) = 2$.

Represent the solution curve and describe its long-term behavior.

$$\varphi'(t) = 2 \cdot 3 \cdot e^{3t} = 6 \cdot e^{3t} = 3 \cdot 2 \cdot e^{3t} = 3 \cdot \varphi(t)$$

$$\varphi(0) = 2 \cdot e^0 = 2 \cdot 1 = 2.$$

$$\Rightarrow \varphi \in S.$$



Real-life ex: money in a bank account (increases exponentially)

Long-term behaviour: $\lim_{t \rightarrow \infty} \varphi(t) = \infty \Rightarrow$ exponentially increasing.
 $\varphi(t) > 0, (\forall) t \in \mathbb{R}.$

1.2. Oscillations with constant amplitude.

$$\varphi: \mathbb{R} \rightarrow \mathbb{R}, \varphi(t) = \frac{1}{6} \sin t, t \in \mathbb{R}.$$

$$\text{Sol. of IVP: } x'' + x = 0, x(0) = 0, x'(0) = \frac{1}{6} ?$$

Repr. solution curve and describe its long-term behavior.

Motion of the simple planar gravity pendulum.

$$\varphi(t) = \eta \cdot \sin t \rightarrow \text{distance between vertical and current position.}$$

$$x' = \varphi'(t) = \eta \cdot \cos t$$

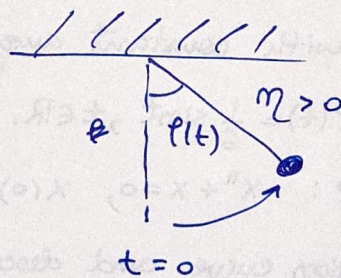
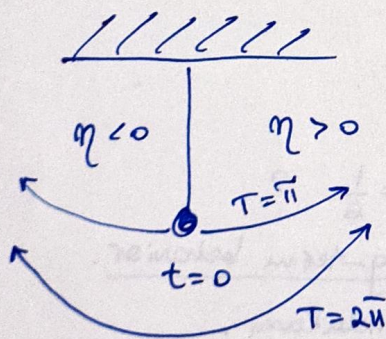
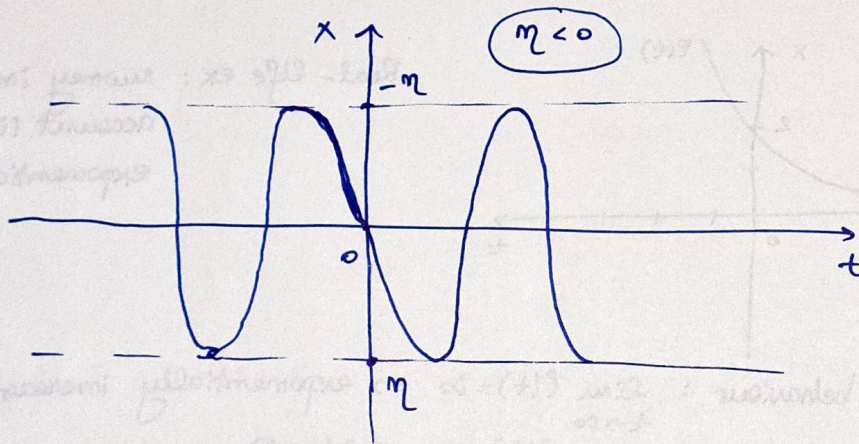
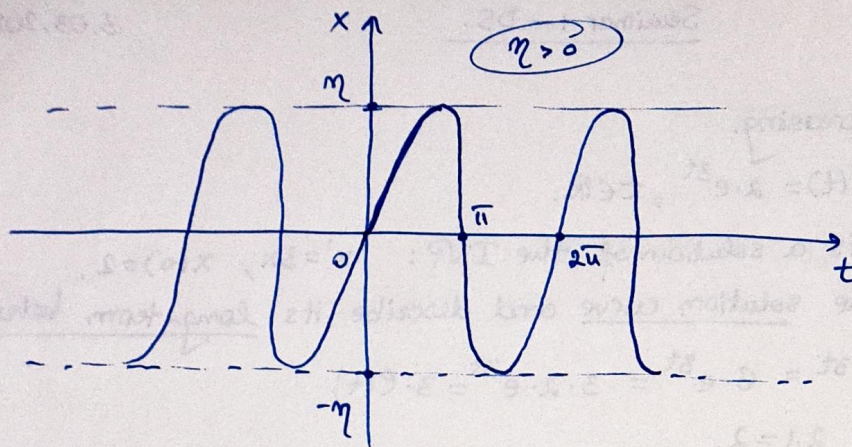
$$\varphi''(t) = -\eta \cdot \sin t$$

$$x'' + x = 0 \Leftrightarrow -\eta \cdot \sin t + \eta \cdot \sin t = 0. \checkmark$$

$$x(0) = \varphi(0) = \eta \cdot \sin 0 = 0 \checkmark$$

$$x'(0) = \varphi'(0) = \eta \cdot \cos 0 = \eta \cdot 1 = \eta \checkmark$$

$$\Rightarrow \varphi \in S.$$



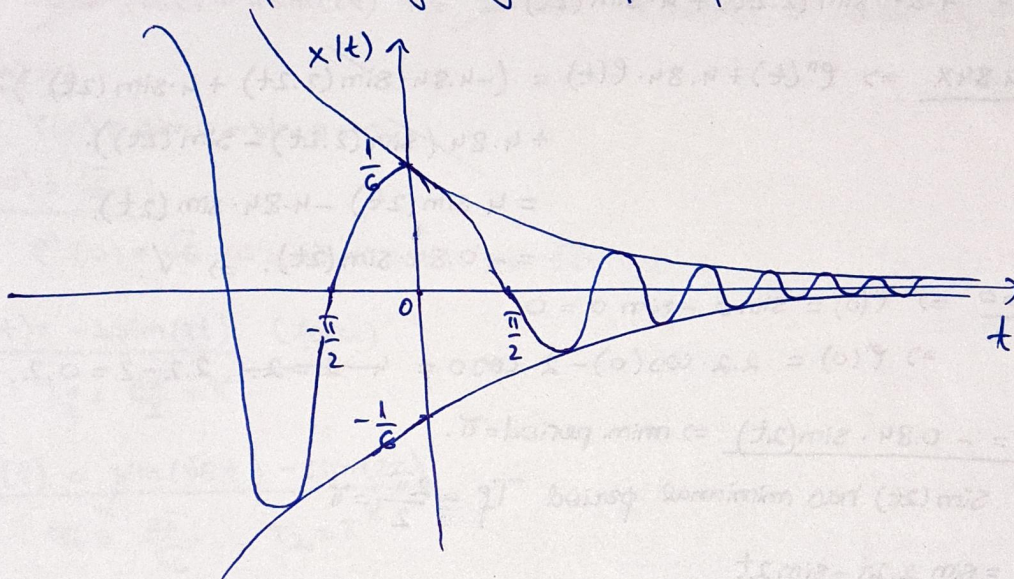
- Long-term behaviour:
- bounded by $-\eta, \eta \rightarrow [-\eta, \eta]$.
 - periodical function with main period $= 2\pi$.
 - oscillates around 0 with constant amplitude $\hookrightarrow |\eta|$.

1.3. Oscillations with exponentially decreasing amplitude.

$$f(t) = \frac{1}{6} \cdot e^{-2t} \cdot \cos t, \quad t \in \mathbb{R}. \quad \text{solution of IVP:}$$

$$x'' + 4x' + 5x = 0, \quad x(0) = \frac{1}{6}, \quad x'(0) = -\frac{1}{3}.$$

- Represent this solution curve and describe its long-term behaviour.
- Motion of the planar gravity damped pendulum.



$$\lim_{t \rightarrow \infty} \frac{1}{6} \cdot \frac{\cos t}{e^{2t}} = 0.$$

$$f(t) = 0 \Leftrightarrow \frac{1}{6} \cdot \frac{\cos t}{e^{2t}} = 0 \Leftrightarrow \cos t = 0 \\ \Rightarrow t = \frac{\pi}{2} + k\pi, \quad k \in \mathbb{Z}.$$

$$\cos(t) = 1 \mid \arccos$$

$$\arccos(\cos(t)) = \arccos 1$$

$$\cos t = 2k\pi.$$

Long-term behaviour: - oscillates around 0 with exponentially decreasing amplitude $\lim_{t \rightarrow \infty} \frac{1}{6} \cdot \frac{\cos t}{e^{2t}} = 0.$

1.4. $\varphi(t) = \sin 2.2t - \sin 2t$, $t \in \mathbb{R}$ sol. of IVP:

$$x'' + 4.84x = -0.84 \cdot \sin 2t, \quad x(0) = 0, \quad x'(0) = 0.2$$

Minimal periods of the force ($f(t) = -0.84 \cdot \sin 2t$) and the response ($\varphi(t) = \sin 2.2t - \sin 2t$) ?

$$\varphi'(t) = 2.2 \cdot \cos(2.2t) - 2 \cdot \cos(2t)$$

$$\varphi''(t) = -4.84 \cdot \sin(2.2t) + 4 \cdot \sin(2t)$$

$$\begin{aligned} \underline{x'' + 4.84x} &\Rightarrow \varphi''(t) + 4.84 \cdot \varphi(t) = (-4.84 \cdot \cancel{\sin(2.2t)} + 4 \cdot \sin(2t)) + \\ &\quad + 4.84(\cancel{\sin(2.2t)} - \sin(2t)). \\ &= 4 \cdot \sin(2t) - 4.84 \cdot \sin(2t). \\ &= -0.84 \cdot \sin(2t). \Rightarrow \checkmark \end{aligned}$$

$$\underline{x(0)=0} \Rightarrow \varphi(0) = \sin 0 - \sin 0 = 0.$$

$$\Rightarrow \varphi'(0) = 2.2 \cdot \cos(0) - 2 \cdot \cos 0 = \cancel{4} - \cancel{2} = 2.2 - 2 = 0.2.$$

$$\underline{f(t) = -0.84 \cdot \sin(2t)} \Rightarrow \text{min. period} = \pi.$$

• $\sin(2t)$ has minimal period $T_f = \frac{2\pi}{2} = \pi$

$$\underline{\varphi(t) = \sin 2.2t - \sin 2t}$$

$$T_1 = \frac{2\pi}{2.2} = \frac{10\pi}{11} \quad \rightarrow \quad T_2 = \frac{2\pi}{2} = \pi$$

• Least common positive period: smallest $T > 0$ s.t. $T = \underbrace{m}_m \cdot \underbrace{\pi}_{T_2} = \underbrace{n}_n \cdot \underbrace{\frac{10\pi}{11}}_{T_1}$.
 $(m, n \in \mathbb{N}). \Rightarrow 11m = 10n.$

$$\Rightarrow \text{smallest solution is } m=10, n=11 \Rightarrow T_p = 10\pi.$$

Conclusion: φ is a solution of the IVP.

minimal periods: $T_f = \pi, T_p = 10\pi.$

$$\Rightarrow f(t), \varphi(t) \rightarrow \text{periodic functions.}$$

1.5) Check $p(t) = \sin \sqrt{6}t - \sin 2t$, $t \in \mathbb{R}$.

$$x'' + 6x = -2 \cdot \sin 2t, \quad x(0) = 0, \quad x'(0) = \sqrt{6} - 2.$$

$$x'' + 6x = -2 \cdot \sin(2t)$$

$$p'(t) = \sqrt{6} \cos(\sqrt{6}t) - 2 \cos(2t)$$

$$p''(t) = -6 \sin(\sqrt{6}t) + 4 \sin(2t)$$

$$-6 \sin(\sqrt{6}t) + 4 \sin(2t) + 6 \cdot \sin \sqrt{6}t - 6 \cdot \sin 2t = -2 \sin 2t \quad \checkmark$$

$$x(0) = 0$$

$$p(0) = \sin 0 - \sin 0 = 0.$$

$$x'(0) = \sqrt{6} - 2$$

$$p'(0) = \sqrt{6} \cdot \cos(0) - 2 \cos(0) = \sqrt{6} - 2 \quad \checkmark$$

$$f(t) = -2 \sin(2t) \quad (\text{force})$$

$$T_f = \frac{2\pi}{2} = \pi$$

$$p(t) = \frac{\sin(\sqrt{6}t)}{T_1} - \frac{\sin(2t)}{T_2}$$

$$T_1 = \frac{2\pi}{\sqrt{6}}, \quad T_2 = \pi$$

$$\text{For } p \text{ to be periodic} \Rightarrow \exists T > 0 \text{ s.t. } T = k \cdot \frac{2\pi}{\sqrt{6}} = m \cdot \pi$$

$$\Rightarrow \mu = \frac{2k}{\sqrt{6}}$$

$\Rightarrow p$ is a solution of the IVP.

$$T_f = \pi$$

$p(t) \rightarrow$ not periodic, but quasiperiodic

1.6) Check $p(t) = t \cdot \sin t$, $t \in \mathbb{R}$, is a solution to IVP.

$$x'' + x = 2 \cdot \cos t, \quad x(0) = 0, \quad x'(0) = 0$$

$$p(t) = \sin t + t \cdot \cos t$$

$$p''(t) = \cos t - \sin t + \cos t - t \cdot \sin t = 2 \cos t - t \cdot \sin t$$

$$p''(t) + p(t) = 2 \cos t - t \cdot \sin t + t \cdot \sin t = 2 \cdot \cos t \quad \checkmark$$

$$p(0) = 0 \cdot \sin 0 = 0 \quad \checkmark$$

$$p'(0) = 0 + 0 \cdot \cos 0 = 0 \quad \checkmark$$

$\Rightarrow p(t) = t \cdot \sin t$ is indeed the solution of the IVP.

Long-term behaviour:

$$-1 \leq \sin t \leq 1 \quad | \cdot t$$

$$-t \leq t \sin t \leq t$$

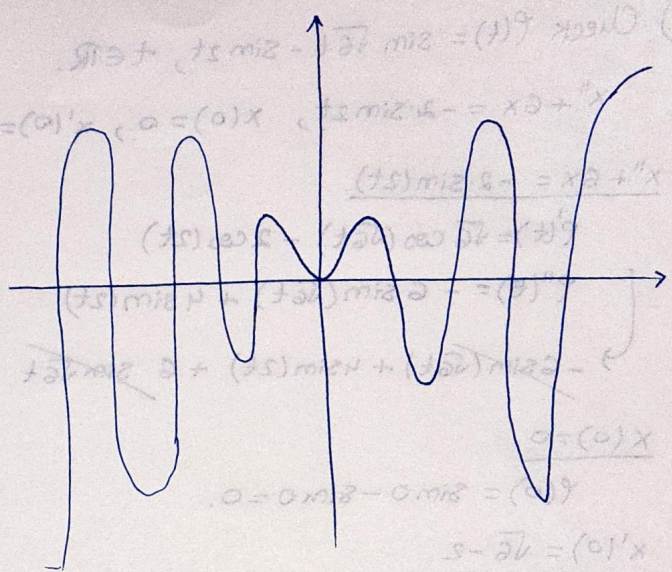
→ oscillates around 0

→ amplitude grows like t

→ unbounded as $t \rightarrow \infty$.

$$t = \frac{\pi}{2} + 2k\pi$$

$$t = \frac{3\pi}{2} + 2k\pi$$



1.7 $y(t) = \cos t$.

$$x' + x = 0, \quad x'' - x = 0, \quad x''' + x' = 0, \quad x^{(4)} + x'' = 0.$$

$$y(t) = -\sin t$$

$$y'(t) = -\cos t$$

$$y''(t) = \sin t$$

$$y^{(4)}(t) = +\cos t$$

$$y'(t) + y(t) = -\sin t + \cos t \neq 0.$$

$$y''(t) - y(t) = -\cos t - \cos t = -2\cos t.$$

$$y'''(t) + y'(t) = \sin t - \sin t = 0. \quad \checkmark$$

$$y^{(4)}(t) + y''(t) = \cos t - \cos t = 0 \quad \checkmark$$

⇒ not a sol.

1.8 Find all solutions of the differential equations.

$$\begin{cases} x(t) = c \\ x'(t) = 0 \\ 0 = f(x) \end{cases}$$

a) $x' = x - x^3$

$$0 = x - x^3 = x(1 - x^2) = x(1 - x)(1 + x)$$

$$\begin{aligned} & \xrightarrow{x=0} \\ & \xrightarrow{x=1} \\ & \xrightarrow{x=-1} \end{aligned}$$

⇒ $x(t) \equiv 0, x(t) \equiv 1, x(t) \equiv -1$ are the constant solutions.

$$b) x' = \sin x.$$

$$0 = \sin x$$

$$\Rightarrow x = m\pi, m \in \mathbb{Z}.$$

$$\Rightarrow x(t) \equiv m\pi \text{ constant solution, } m \in \mathbb{Z}.$$

$$c) x' = \frac{x+1}{2x^2+5}$$

$$0 = \frac{x+1}{2x^2+5} \Rightarrow x = -1 \Rightarrow x(t) \equiv -1 \text{ ct. sol.}$$

$$d) x' = x^2 + x + 1$$

$$0 = x^2 + x + 1$$

$$\Delta = 1 - 4 = -3 < 0 \Rightarrow \text{no real ct. solutions.}$$

$$e) x' = x + 4x^3$$

$$0 = x + 4x^3 = x(1 + 4x^2) \begin{cases} \nearrow x = 0 \\ \searrow - \end{cases}$$

$$\Rightarrow x(t) \equiv 0 \text{ ct. sol.}$$

$$f) x' = -1 + x + 4x^3.$$

$$0 = -1 + x + 4x^3.$$

$$x = \frac{1}{2}: 4 \cdot \left(\frac{1}{2}\right)^3 + \frac{1}{2} - 1 = 1 \cdot \frac{1}{8} + \frac{1}{2} - 1 = -1 = 0 \checkmark.$$

root.

$$4x^3 + x - 1 = (2x-1)(2x^2 + x + 1).$$

$$\Delta = 1 - 8 = -7 < 0. \text{ no real roots.}$$

$$\Rightarrow x(t) \equiv \frac{1}{2} \text{ ct. sol.}$$

1.9. i) $x_1, x_2, x_3: \mathbb{R} \rightarrow \mathbb{R}$; Prove that x_1, x_2, x_3 are lin. ind. in the linear space $C(\mathbb{R})$.

$$? \rightarrow x_1(t) = 1, x_2(t) = t, x_3(t) = t^2, (\forall) t \in \mathbb{R}.$$

$$\boxed{\alpha \cdot x_1(t) + \beta \cdot x_2(t) + \gamma \cdot x_3(t) = 0, (\forall) t \in \mathbb{R}.}$$

$$\Rightarrow \alpha \cdot 1 + \beta \cdot t + \gamma \cdot t^2 = 0, (\forall) t \in \mathbb{R}.$$

$$\Rightarrow \begin{cases} \alpha = 0 \\ \beta = 0 \\ \gamma = 0 \end{cases} \Rightarrow \{1, t, t^2\} \text{ linearly indep. in } C(\mathbb{R}).$$

$$ii) x(t) = at^2 + bt + c, \quad a, b, c \in \mathbb{R}.$$

$$\downarrow x' - 5x = 2t^2 + 3.$$

$$x'(t) = 2 \cdot at + b.$$

$$2 \cdot at + b - 5(at^2 + bt + c) = 2t^2 + 3.$$

$$\Rightarrow -5at^2 + (2a - 5b)t + (b - 5c) = 2t^2 + 3.$$

$$\Rightarrow \begin{cases} -5a = 2 \Rightarrow a = -\frac{2}{5} \\ 2a - 5b = 0 \Rightarrow 2 \cdot -\frac{2}{5} - 5b = 0 \Rightarrow 5b = -\frac{4}{5} \Rightarrow b = -\frac{4}{25} \\ b - 5c = 3 \Rightarrow b = 3 + 5c = 3 + 5 \cdot -\frac{4}{25} = 3 - \frac{4}{5} \end{cases}$$

$$\Rightarrow 5c = b - 3 = -\frac{4}{25} - 3 = -\frac{79}{25} \Rightarrow c = -\frac{79}{125}.$$

$$x(t) = -\frac{2}{5}t^2 - \frac{4}{25}t - \frac{79}{125}.$$

$$(1.b). i) x_1, x_2, x_3: \mathbb{R} \rightarrow \mathbb{R}$$

$$x_1(t) = \cos t, \quad x_2(t) = \sin t, \quad x_3(t) = e^t.$$

$$\boxed{a \cdot x_1(t) + b \cdot x_2(t) + c \cdot x_3(t) = 0}$$

$$\Rightarrow a \cdot \cos t + b \cdot \sin t + c \cdot e^t = 0. \quad (1)$$

$$t=0: a \cdot \cos 0 + b \cdot \sin 0 + c \cdot e^0 = 0.$$

$$\Rightarrow a + b + c = 0$$

$$\Rightarrow \underline{a + c = 0}.$$

$$(1)' \Rightarrow -a \cdot \sin t + b \cdot \cos t + c \cdot e^t = 0 \quad (2)$$

$$t=0: -a \cdot \sin 0 + b \cdot \cos 0 + c \cdot e^0 = 0$$

$$\Rightarrow \underline{b + c = 0}.$$

$$(2)' \Rightarrow -a \cdot \cos t - b \cdot \sin t + c \cdot e^t = 0$$

$$t=0: -a \cdot \cos 0 - b \cdot \sin 0 + c \cdot e^0 = 0$$

$$\Rightarrow \underline{-a + c = 0}.$$

$$\begin{cases} a + c = 0 \\ b + c = 0 \\ -a + c = 0 \end{cases} \Rightarrow \begin{cases} 2c = 0 \Rightarrow c = 0; a = 0; b = 0 \end{cases}$$

$\Rightarrow \cos t, \sin t, e^t$ are lin. indep. in $C(\mathbb{R})$.

1.10 ii) $x(t) = a \cdot \cos t + b \cdot \sin t + c \cdot e^t$, $a, b, c \in \mathbb{R} \rightarrow$ solution ... $x = e^t + \dots$ (b)

... of $x''' + x = -3 \sin t + 2 \cdot e^t$.

~~$x'(t) = -a \sin t + b \cos t + c \cdot e^t$~~

$x'(t) = -a \sin t + b \cos t + c \cdot e^t$

$x''(t) = -a \cos t - b \sin t + c \cdot e^t$

$x'''(t) = a \sin t - b \cos t + c \cdot e^t$

$x''' + x = a \sin t - b \cos t + c \cdot e^t + a \cos t + b \sin t + c \cdot e^t$
 $= (a-b) \cos t + (a+b) \sin t + 2 \cdot c \cdot e^t$

$(a-b) \cos t + (a+b) \sin t + 2 \cdot c \cdot e^t = -3 \sin t + 2 \cdot e^t$

$\Rightarrow \begin{cases} a-b=0 \\ a+b=-3 \\ 2 \cdot c=2 \end{cases} \Rightarrow \begin{cases} 2a=-3 \Rightarrow a=-\frac{3}{2} \\ b=a=-\frac{3}{2} \\ c=1 \end{cases}$

$x(t) = -\frac{3}{2} \cos t - \frac{3}{2} \sin t + e^t$

1.11. $x(t) = e^{kt}$, $k \in \mathbb{R}$.

a) $x'' - 5x' + 6x = 0 \Leftrightarrow k^2 \cdot e^{kt} - 5 \cdot k \cdot e^{kt} + 6 \cdot e^{kt} = 0 \mid : e^{kt} \neq 0$

$x'(t) = k \cdot e^{kt} \Leftrightarrow k^2 - 5k + 6 = 0$

$x''(t) = k^2 \cdot e^{kt} \Leftrightarrow (k-2)(k-3) = 0$

$x'''(t) = k^3 \cdot e^{kt} \Rightarrow k=2, k=3$

$x^{(4)}(t) = k^4 \cdot e^{kt}$

$\Rightarrow x(t) = e^{2t}$
 $\Rightarrow x(t) = e^{3t}$ } exponential solutions.

b) $x''' - 5x'' + 6x' = 0$

$\Leftrightarrow k^3 \cdot e^{kt} - 5 \cdot k^2 \cdot e^{kt} + 6 \cdot k \cdot e^{kt} = 0 \mid : e^{kt}$

$\Leftrightarrow k^3 - 5k^2 + 6k = 0$

$\Rightarrow k(k^2 - 5k + 6) = k(k-2)(k-3) = 0$

$\begin{cases} k=0 \Rightarrow e^{0t}=1 \\ k=2 \Rightarrow e^{2t} \\ k=3 \Rightarrow e^{3t} \end{cases}$

c) $x^{(4)} - 5x''' + 6x'' = 0$

$\Leftrightarrow k^4 \cdot e^{kt} - 5 \cdot k^3 \cdot e^{kt} + 6 \cdot k^2 \cdot e^{kt} = 0 \mid : e^{kt}$

$\Leftrightarrow k^4 - 5k^3 + 6k^2 = 0$

$\Rightarrow k^2(k^2 - 5k + 6) = 0$

$\Rightarrow k^2(k-2)(k-3) = 0$

$\begin{cases} k_1 = k_2 = 0 \Rightarrow e^{0t} = 1 \\ k_3 = 2 \Rightarrow e^{2t} \\ k_4 = 3 \Rightarrow e^{3t} \end{cases}$

$$d) x'' + 9x = 0$$

$$\Rightarrow x^2 \cdot e^{xt} + 9 \cdot e^{xt} = 0 \quad | : e^{xt}$$

$$\Rightarrow x^2 + 9 = 0$$

$$\Rightarrow x^2 = -9 \Rightarrow \text{no real sol.}$$

$$\Rightarrow \text{no sol. of the form } e^{xt} \text{ with } x \in \mathbb{R}$$

$$e) t^2 x'' - tx' + x = 0$$

$$\Rightarrow t^2 \cdot x^2 \cdot e^{xt} - t \cdot x \cdot e^{xt} + e^{xt} = 0 \quad | : e^{xt}$$

$$\Rightarrow t^2 \cdot x^2 - t \cdot x + 1 = 0$$

$$\Rightarrow \Delta = x^2 - 4 \cdot x^2 = -3x^2 < 0 \Rightarrow \text{no sol. real.}$$

1.13 Integrate.

$$a) x' = 0 \Rightarrow x(t) = C$$

$$b) x' = 2t$$

$$\int 2t dt = 2 \cdot \frac{t^2}{2} + C = t^2 + C = x(t)$$

$$c) x' = \sin t$$

$$\int \sin t dt = -\cos t + C = x(t)$$

$$d) x' = 2t + \sin t$$

$$\int (2t + \sin t) dt = t^2 - \cos t + C = x(t)$$

$$e) x' = e^{2t} \cdot \cos t$$

$$I = \int (e^{2t} \cdot \cos t) dt = \frac{1}{2} \int (e^{2t})' \cdot \cos t dt = \frac{1}{2} \cdot (e^{2t} \cdot \cos t + \int e^{2t} \cdot \sin t dt)$$

$$\int e^{2t} \cdot (\sin t)' dt = e^{2t} \cdot \sin t - \int e^{2t} \cdot \sin t dt$$

$$\int e^{2t} \cdot \sin t dt = \frac{1}{2} \int (e^{2t})' \cdot \sin t dt = \frac{1}{2} (e^{2t} \cdot \sin t - \int e^{2t} \cdot \cos t dt)$$

$$u = \sin t \quad |' = du = \cos t dt$$

$$dv = e^{2t} dt; v = \frac{1}{2} e^{2t}$$

$$I = \frac{1}{2} e^{2t} \cdot \cos t + \frac{1}{2} \int e^{2t} \cdot \sin t dt + \frac{1}{2} e^{2t} \cdot \sin t - \frac{1}{2} \int e^{2t} \cdot \cos t dt$$

$$= \frac{1}{2} e^{2t} \cdot \cos t + \frac{1}{2} \left(\frac{1}{2} e^{2t} \cdot \sin t - \frac{1}{2} \int e^{2t} \cdot \cos t dt \right)$$

$$= \frac{1}{2} e^{2t} \cdot \cos t + \frac{1}{4} e^{2t} \cdot \sin t - \frac{1}{4} I \Rightarrow 1 + \frac{1}{4} I = \frac{1}{2} e^{2t} \left(\cos t + \frac{1}{2} \sin t \right)$$

$$\Rightarrow \frac{5}{4} I = \frac{1}{2} e^{2t} \cdot \left(\cos t + \frac{1}{2} \sin t \right) \quad | \cdot 2 \Rightarrow$$

$$\Rightarrow \frac{5}{4} I = e^{2t} \cdot \frac{1}{4} \cdot e^{2t} (2 \cos t + \sin t)$$

$$\Rightarrow I = \frac{1}{5} \cdot \frac{1}{4} \cdot e^{2t} (2 \cos t + \sin t) + C$$

$$\Rightarrow I = \frac{1}{5} \cdot e^{2t} \cdot (2 \cos t + \sin t) + C = x(t).$$

$$f) x' = (t^2 - 5t + 7) \cdot \sin t.$$

$$I = \int (t^2 - 5t + 7) \cdot \sin t \, dt = \int (t^2 - 5t + 7) (\cos t)' \, dt =$$

$$= - \int (t^2 - 5t + 7) \cdot \cos t \, dt + \int (2t - 5) \cdot \cos t \, dt$$

$$= (-t^2 + 5t - 5) \cos t + (2t - 5) \sin t + C.$$

$$\int (2t - 5) \cdot \cos t \, dt = \int (2t - 5) \cdot (\sin t)' \, dt = (2t - 5) \sin t - \int 2 \cdot \sin t \, dt$$

$$= (2t - 5) \sin t - 2 \cdot (-\cos t) + C = (2t - 5) \sin t + 2 \cos t + C.$$

$$\int (t^2 - 5t + 7) \cdot \cos t \, dt$$

$$g) x' = e^{t^2}$$

$$x(t) = \int_0^t e^{s^2} \, ds + C$$

$$h) x'' = -3.$$

$$\int -3 \, dt = -3t + C_1$$

$$\int -3t \, dt = -3 \cdot \frac{t^2}{2} + C_1$$

$$\int (-3t + C_1) \, dt = -3 \cdot \frac{t^2}{2} + t \cdot C_1 + C_2$$

$$i) x''' = 0$$

$$\int 0 \, dt = C_1$$

$$\int C_1 \, dt = C_1 t + C_2$$

$$\int (C_1 t + C_2) \, dt = C_1 \cdot \frac{t^2}{2} + C_2 t + C_3$$

$$x(t) = A t^2 + B t + C. \quad (A, B, C \rightarrow \text{const.})$$

$$j) tx' + x = 0.$$

$$? (tx)' = t \cdot x' + x. \Rightarrow (tx)' = 0 \Rightarrow tx = C.$$

$$\Rightarrow x(t) = \frac{C}{t}, t \neq 0.$$

$$\int \frac{1}{t} dt = \ln|t| + C$$

$$k) tx' + x = 1.$$

$$? (tx)' = 1 \Rightarrow \int (tx)' dt = \int 1 dt \Rightarrow tx = t + C.$$

$$(tx)' = t \cdot x' + t \cdot x' = 1.$$

$$l) t^3 x' + 3t^2 x = 1$$

$$(t^3 \cdot x)' = t^3 \cdot x' + 3t^2 \cdot x = 1.$$

$$\int t^3 \cdot x$$

$$t^3 \cdot x' + 3t^2 \cdot x = 1 \Rightarrow (t^3 x)' = 1 \Rightarrow t^3 x = t + C \Rightarrow x = \frac{t+C}{t^3}$$

$$t^3 \cdot x' + 3t^2 \cdot x = 1 \Rightarrow t^3 \cdot x' + 3t^2 \cdot x = 1 \Rightarrow t^3 \cdot x' + 3t^2 \cdot x = 1$$

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1.14. General sol. of $x' = ax$.

$x = c \cdot e^{at}$ describes the set of all sol. of diff. eq. $x' = ax$. (1)

$$x(t) = c \cdot e^{at}$$

$$x'(t) = c \cdot a \cdot e^{at} = a \cdot (c \cdot e^{at}) = a \cdot x(t). \checkmark$$

$\Rightarrow$ (1) function of the form $c \cdot e^{at}$ satisfies diff. eq. (1)

Let $x(t)$ be ⁽²⁾ sol. of $x' = ax$

$$\text{Define } y(t) = x(t) \cdot e^{-at}$$

$$y'(t) = x'(t) \cdot e^{-at} + x(t) \cdot (-a) \cdot e^{-at} \\ = x'(t) \cdot e^{-at} - a \cdot x(t) \cdot e^{-at}$$

$$\text{But } x' = ax$$

$$\Rightarrow y'(t) = ax \cdot e^{-at} - ax \cdot e^{-at} = 0. \Rightarrow y \text{ is constant.}$$

$$= ax \cdot e^{-at} \cdot (1 - 1)$$

$$\Rightarrow \exists c \in \mathbb{R} \text{ s.t. } y(t) = c.$$

$$\Rightarrow x(t) \cdot e^{-at} = c \cdot e^{at}$$

$$\Rightarrow x(t) = c \cdot e^{at}. \Rightarrow \text{(1) sol. has this form.}$$

1.15. $x'' = a^2 x$

$$x = c_1 \cdot e^{at} + c_2 \cdot e^{-at}; c_1, c_2 \in \mathbb{R}.$$

$$x'(t) = c_1 \cdot a \cdot e^{at} + c_2 \cdot (-a) \cdot e^{-at}.$$

$$x''(t) = c_1 \cdot a^2 \cdot e^{at} + c_2 \cdot a^2 \cdot e^{-at} = a^2 \cdot (c_1 \cdot e^{at} + c_2 \cdot e^{-at}) \checkmark$$

Let $x(t)$ be ⁽²⁾ sol. of $x'' = a^2 x$.

$$\text{Define } y = x' - ax; \Rightarrow y(t) = x'(t) - a \cdot x(t).$$

$$y'(t) = x''(t) - a \cdot x'(t) = a^2(c_1 \cdot e^{at} + c_2 \cdot e^{-at}) - \\ - a \cdot (c_1 \cdot a \cdot e^{at} + c_2 \cdot (-a) \cdot e^{-at})$$

$$= a^2 \cdot c_1 \cdot e^{at} + a^2 \cdot c_2 \cdot e^{-at} - a^2 \cdot c_1 \cdot e^{at} + a^2 \cdot c_2 \cdot e^{-at} \\ = 2 \cdot a^2 \cdot c_2 \cdot e^{-at}$$

$$=$$

(1.6) $x'' = -x$

$x = c_1 \cdot \cos t + c_2 \cdot \sin t$; $c_1, c_2 \in \mathbb{R}$.

$x'' = -x$.

$x'(t) = c_1 \cdot (-\sin t) + c_2 \cdot \cos t$.

$x''(t) = -c_1 \cos t - c_2 \sin t = -(c_1 \cos t + c_2 \sin t)$.

$y = x \cdot \cos t - x' \cdot \sin t$.

$z = x' \cdot \cos t + x \cdot \sin t$.

$y'(t) = x'(t) \cdot \cos t + x(t) \cdot (-\sin t) - (x''(t) \cdot \sin t + x'(t) \cdot \cos t)$

$y'(t) = -x(t) \cdot \sin t - x''(t) \cdot \sin t$

$= -\sin t (x(t) + x''(t))$

$= 0$.

$z'(t) = x''(t) \cdot \cos t + x'(t) \cdot (-\sin t) + x'(t) \cdot \sin t + x(t) \cdot \cos t$
 $= \cos t (x''(t) + x(t))$

(1.7) $x'' = -x$

$x = c_1 e^{it} + c_2 e^{-it}$; $c_1, c_2 \in \mathbb{C}$

$x'(t) = c_1 i e^{it} - c_2 i e^{-it}$

$x''(t) = c_1 i^2 e^{it} + c_2 i^2 e^{-it} = -c_1 e^{it} - c_2 e^{-it} = -x$

Let $x(t)$ be sol. of $x'' = -x$.

Define $y = x' - ix$; $y' = x'' - i x' = -x - i x' = -(x' + i x)$

$y'(t) = x''(t) - i x'(t) = -x(t) - i x'(t)$

$y(t) = (c_1 e^{it} + c_2 e^{-it}) - i(c_1 e^{it} - c_2 e^{-it})$

$y(t) = c_1 e^{it} + c_2 e^{-it} - i c_1 e^{it} + i c_2 e^{-it}$

$y(t) = c_1 (1 - i) e^{it} + c_2 (1 + i) e^{-it}$